

Basic Principles of Medicine 1

Module : Musculoskeletal System

MSK DLA

DLA Title: Genetics Cases OI and DMD

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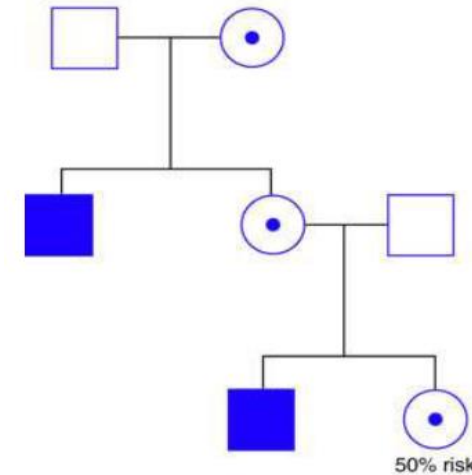
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Sobering, PhD
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Learning Objectives:

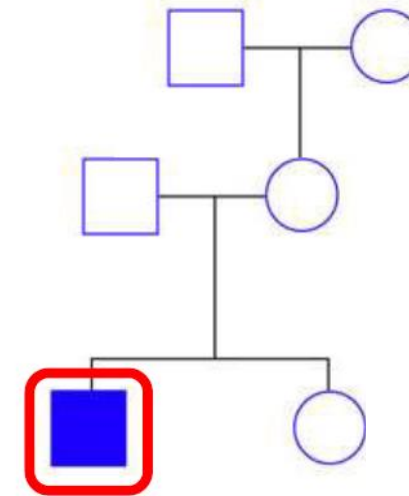
SOM.MK.III.BPM1.1.MSK.1.GNET.0156	Discuss elevated creatine kinase levels in DMD and BMD patients
SOM.MK.III.BPM1.1.MSK.1.GNET.0157	Describe the inheritance pattern of Duchenne muscular dystrophy and discuss the clinical features
SOM.MK.III.BPM1.1.MSK.1.GNET.0158	Compare and contrast the allelic heterogeneity in the DMD gene that leads to either Becker muscular dystrophy or Duchenne muscular dystrophy
SOM.MK.III.BPM1.1.MSK.1.GNET.0159	Analyze the DMD gene and discuss why it is sometimes a hotspot for mutation in the human genome
SOM.MK.III.BPM1.1.MSK.1.GNET.0160	Discuss de novo mutations which lead to 1/3 of all cases of DMD
SOM.MK.III.BPM1.1.MSK.1.GNET.0161	Explain the effects of skewed X-inactivation leading to a manifesting heterozygote in females
SOM.MK.III.BPM1.1.MSK.1.GNET.0162	Describe the structure of collagen
SOM.MK.III.BPM1.1.MSK.1.GNET.0163	Describe the general features of osteogenesis imperfecta
SOM.MK.III.BPM1.1.MSK.1.GNET.0164	Compare and contrast how haploinsufficiency might be used to explain one type of OI, and dominant negative might be used to describe a different form of this disorder
SOM.MK.III.BPM1.1.MSK.1.GNET.0165	Describe classic non-deforming OI with blue sclera (formally known as OI type 1) using the haploinsufficiency model. Describe perinatally lethal osteogenesis imperfecta (formerly known as OI type 2) using the dominant negative model.

Duchenne Muscular Dystrophy (Becker Muscular Dystrophy):

- Both X-linked recessive disorders associated with mutations in gene encoding Dystrophin.
- DMD approx. frequency 1/5000 males, resulting from mutations and deletions leading to **out-of-frame exons**.
- Genetic fitness of affected males = 0
- 1/3 of new cases due to de novo mutation (sperm or oocyte formation)
- Becker Muscular Dystrophy (BMD) approx. frequency 1/18000 males, resulting from mutations and deletions leading to **in-frame exons**.



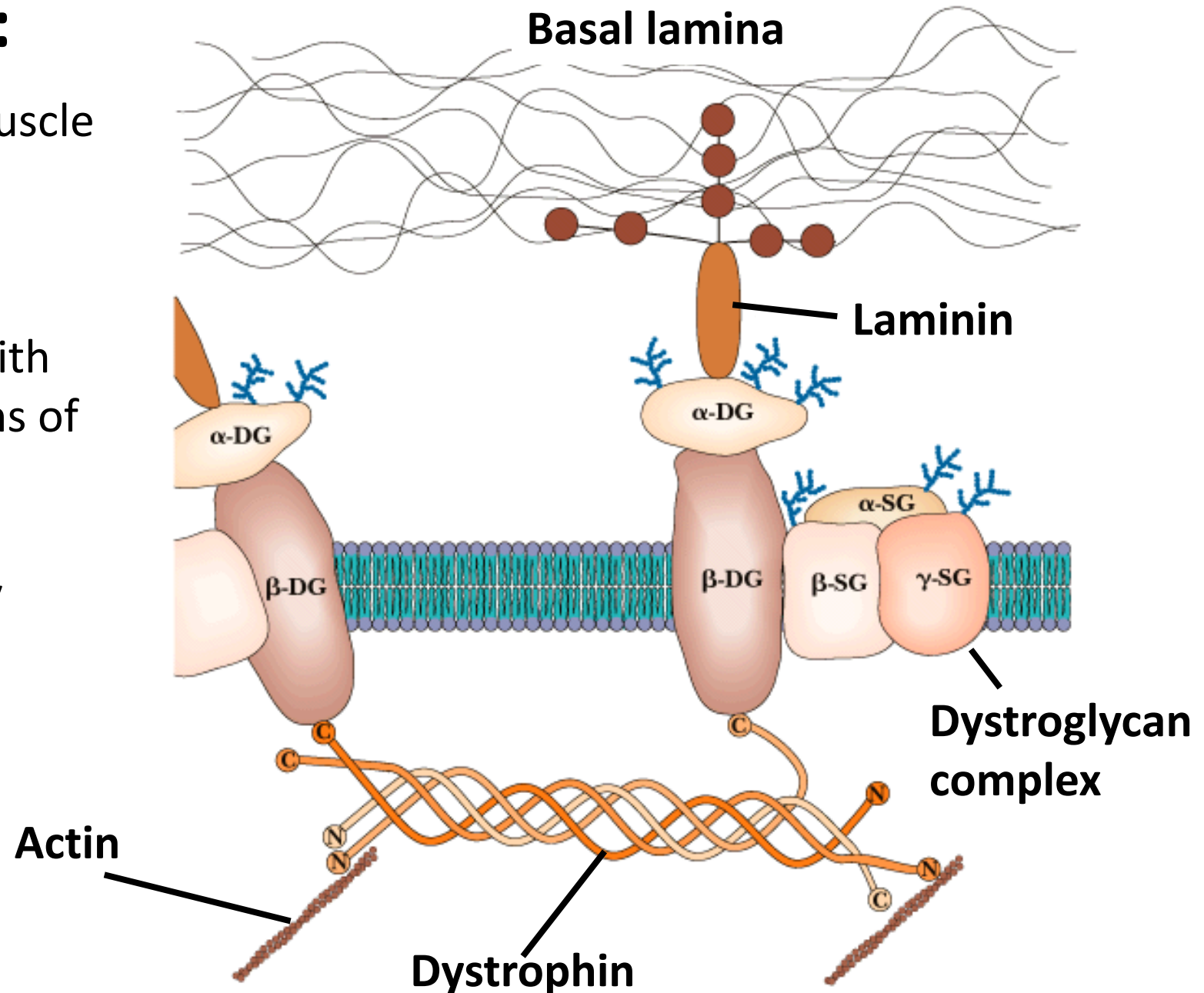
X-linked recessive inheritance pattern



De novo mutation inheritance pattern

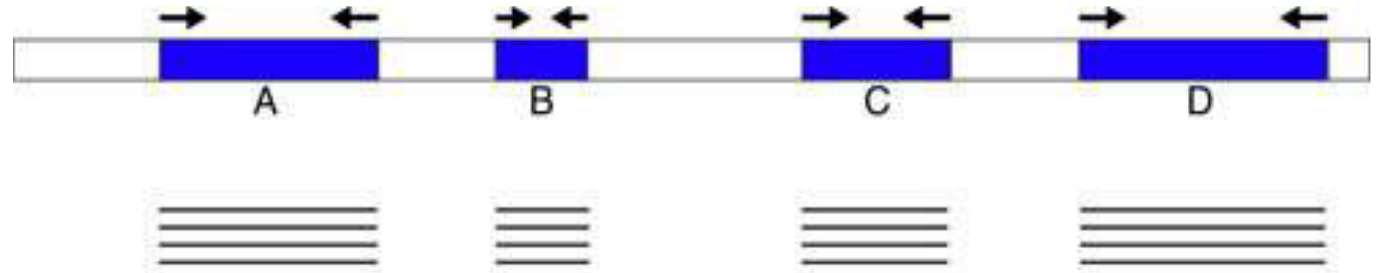
Function of Dystrophin:

- Anchors cytoskeleton (actin) of muscle cells to extracellular matrix (basal lamina)
- Links actin filaments associated with sarcolemma membrane to proteins of muscle cell membrane
- Essential for muscle fiber integrity

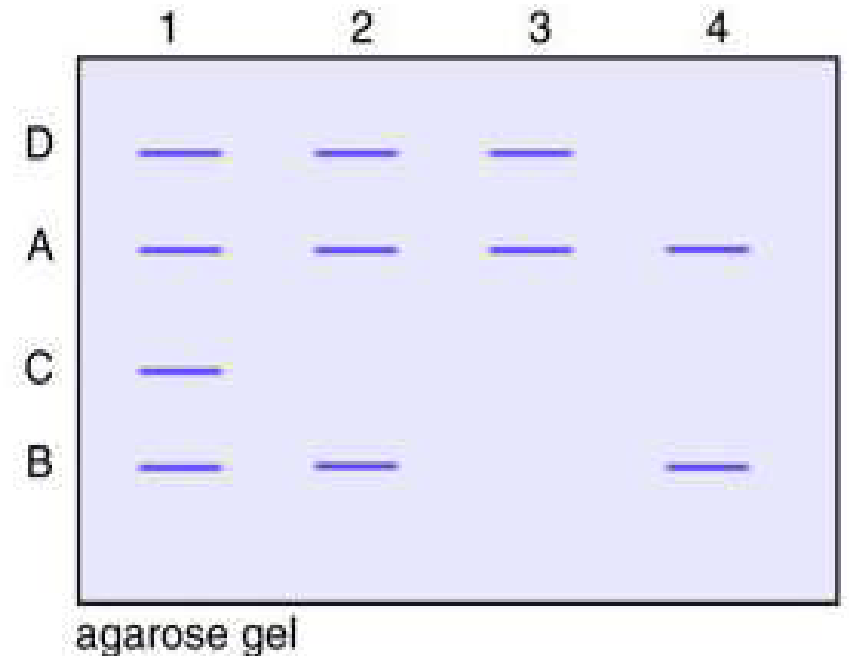


Diagnosis of DMD:

- Dystrophin – 3 domain structure – actin binding, repeat and ECM binding domains → big protein = big gene (79 exons/2.3 Mb)!



- Mutations (mainly large deletions), detected using numerous multiplex PCR assays in tandem
- Deletion = loss/size reduction of exon on agarose gel electrophoresis
- Other mutations/gene duplication leading to DMD detected by direct sequencing



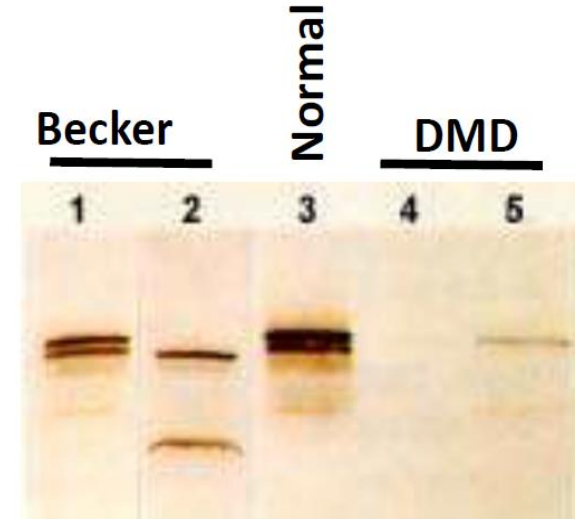
Clinical and Lab Findings:

- DMD – loss of expression of Dystrophin protein.
- BMD – reduction in expression or defective Dystrophin protein.

Clinical Features:

- Degeneration skeletal muscle cells – atrophy and weakness
- ↑ creatine kinase (CK-MM) in both disorders
- Replacement of muscle with yellowish/white fat
- Cardiomyopathy and cognitive dysfunction
- DMD – early age onset, wheelchair-bound in teens
- Death usually before reaching 20s; Genetic fitness = 0
- BMD – later onset and slower rate of progression
- Severity of symptoms variable
- BMD – approx. 90% mutations inherited

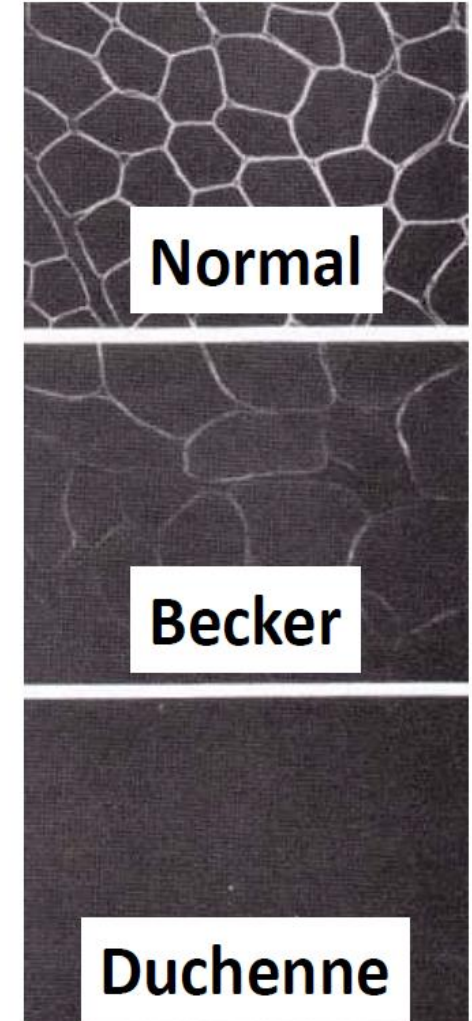
Western Blot Analysis



DMD Muscle Biopsy

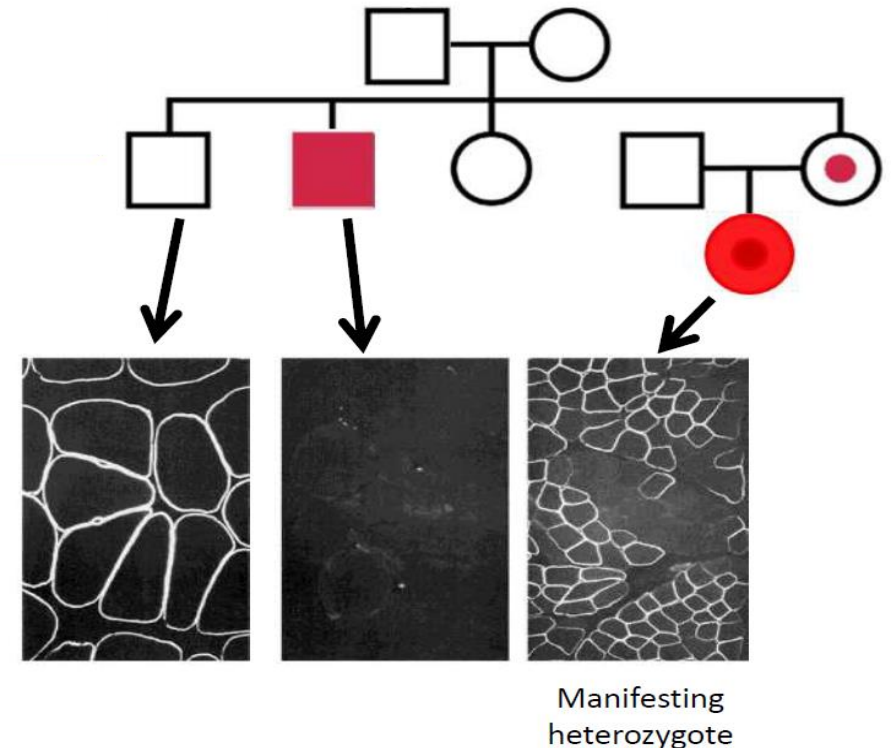
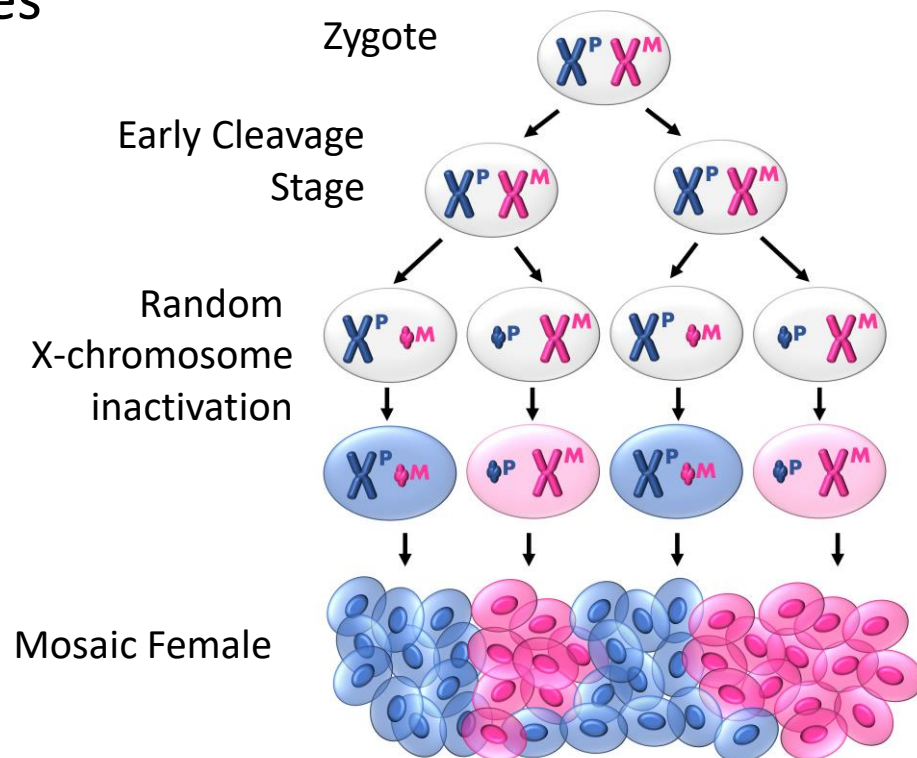


Immunofluorescence



Carrier Females:

- Female DMD carriers ↑ risk of adult-onset cardiology events
- Skewed X-inactivation early in development leads to mosaicism in adult tissues (Mutant X active / Normal X inactivated = No functional protein)
- Some tissues healthy others badly affected – observation in DMD symptoms in heterozygote females



Summary and Genetic Principles Illustrated:

Duchenne Muscular Dystrophy/ Becker muscular dystrophy:

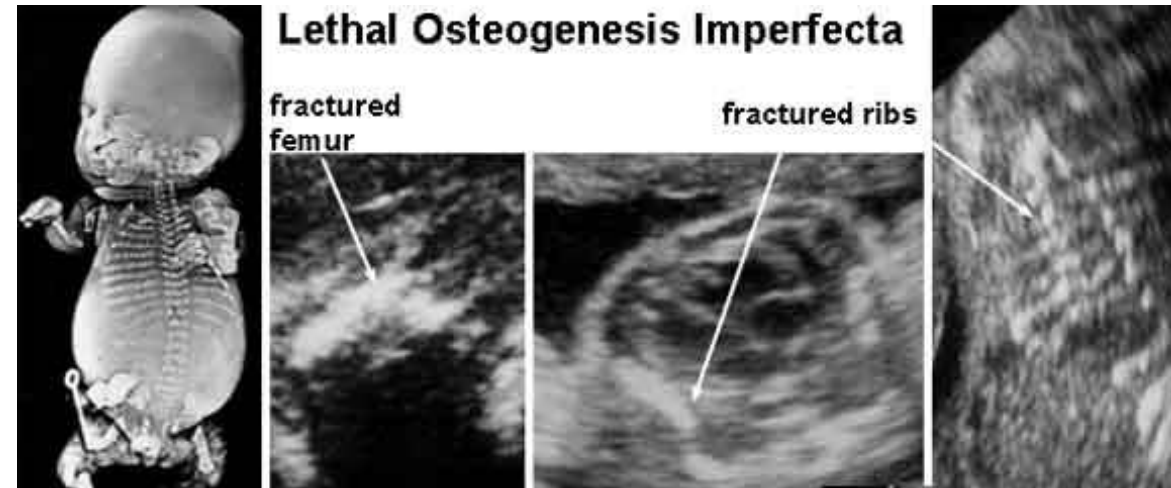
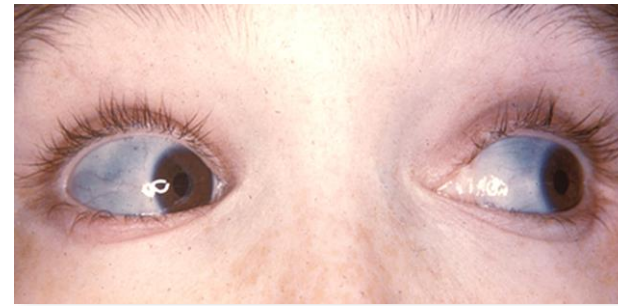
- X-linked recessive inheritance
- High mutation rate
- Allelic heterogeneity
- X-inactivation and carrier risk
- Variable expressivity
 - Severe in DMD
 - Less severe in BMD

Osteogenesis Imperfecta:

- Autosomal dominant disorder
- Mutation - *COL1A1* and *COL1A2* genes
- Type I to VIII (Type I - mild, type II - most severe)
- "Brittle bone disease" – bones easily bend/fracture

Clinical Features:

- Mild forms (Classic Non-deforming):
 - Bone fractures due to minor trauma
 - Blue or grey tint to sclera.
 - Hearing loss in adulthood.
 - Delayed wound healing
 - Spine Deformity (kyphosis)
- Severe form (Perinatally lethal): Death in utero or shortly after birth from pulmonary hypoplasia



Collagen Structure:

- Characteristic repetitive primary structure rich in proline and lysine.

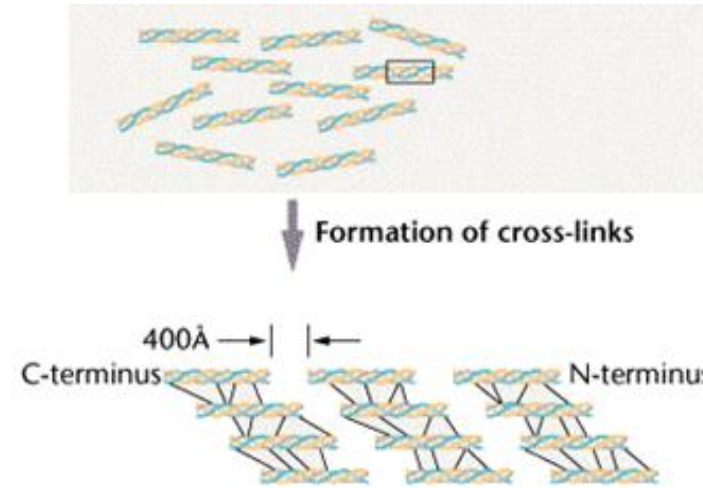
Conserved amino acid Sequence

$\text{-Gly-X-Y-Gly-X-Y-Gly-X-Y-}$

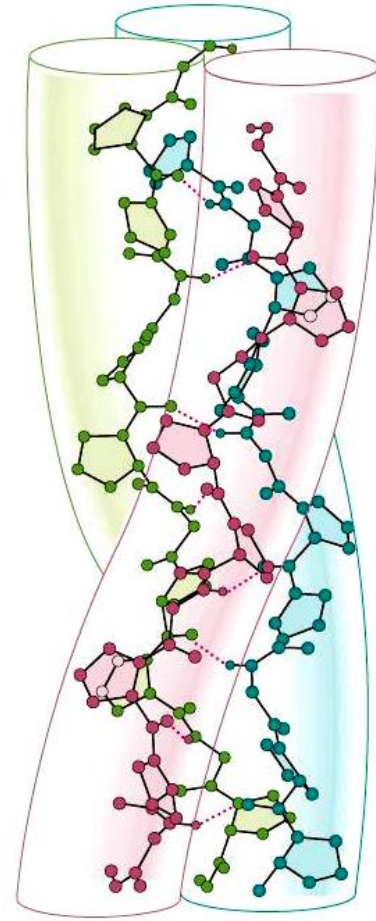
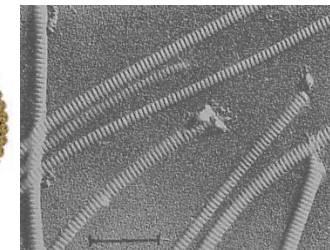
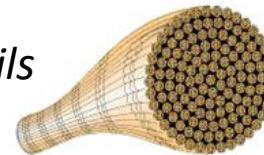
X = frequently proline

Y = proline or lysine

- Pro-alpha chains held together via hydrogen bonding – hydroxyproline (prolyhydroxylase; Vit C).
- Secreted as tropocollagen – collagen fibrils via process of **Fibrillogenesis**.
- Fiber crosslinking – lysyl oxidase (Cu^{2+}) via allysine formation.
- Gaps – site of bone mineralization (glycosylated hydroxylysine).



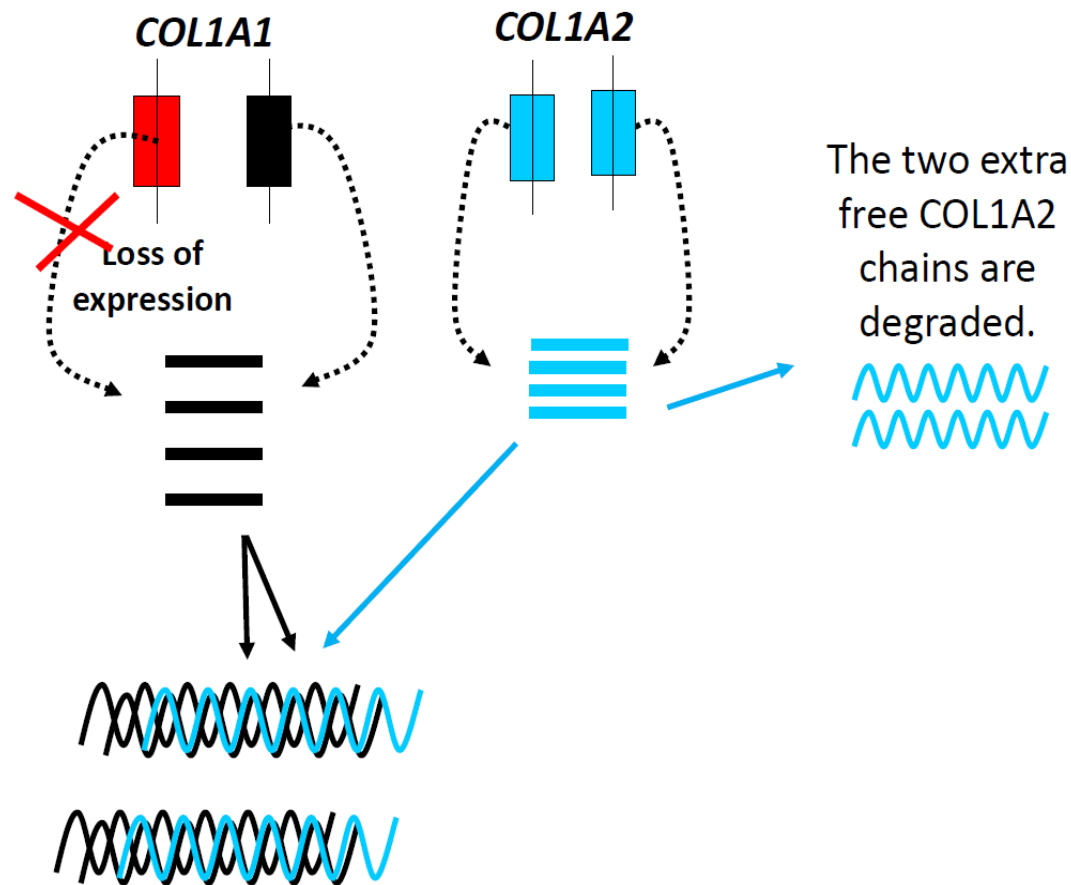
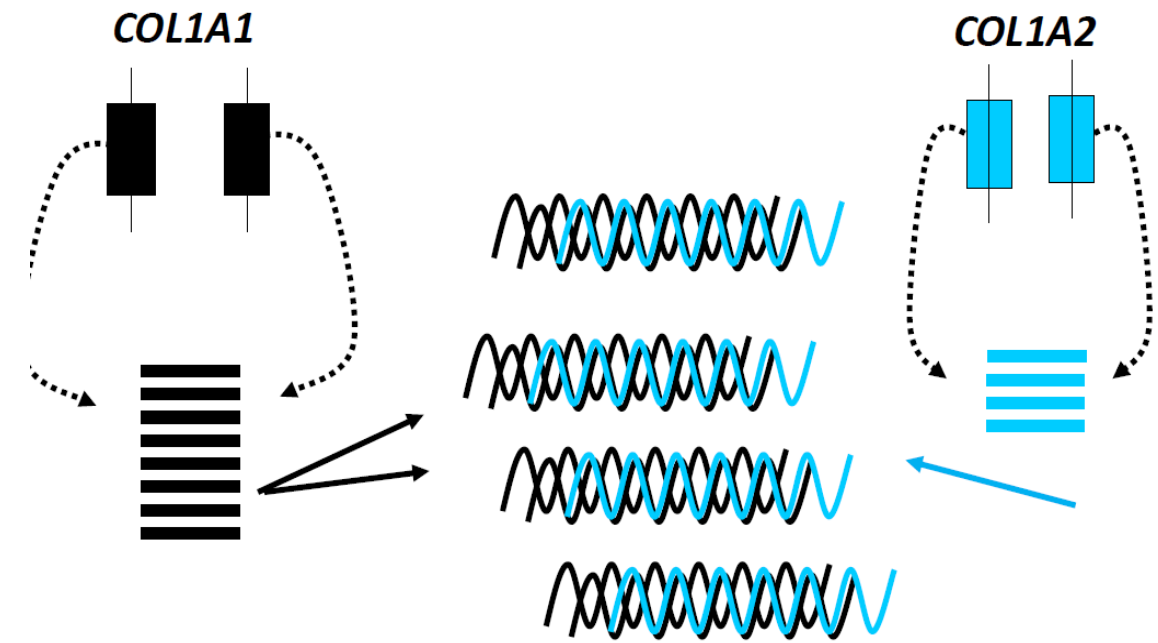
Collagen Fibrils



Triple helix/superhelix

Haploinsufficiency (Type I):

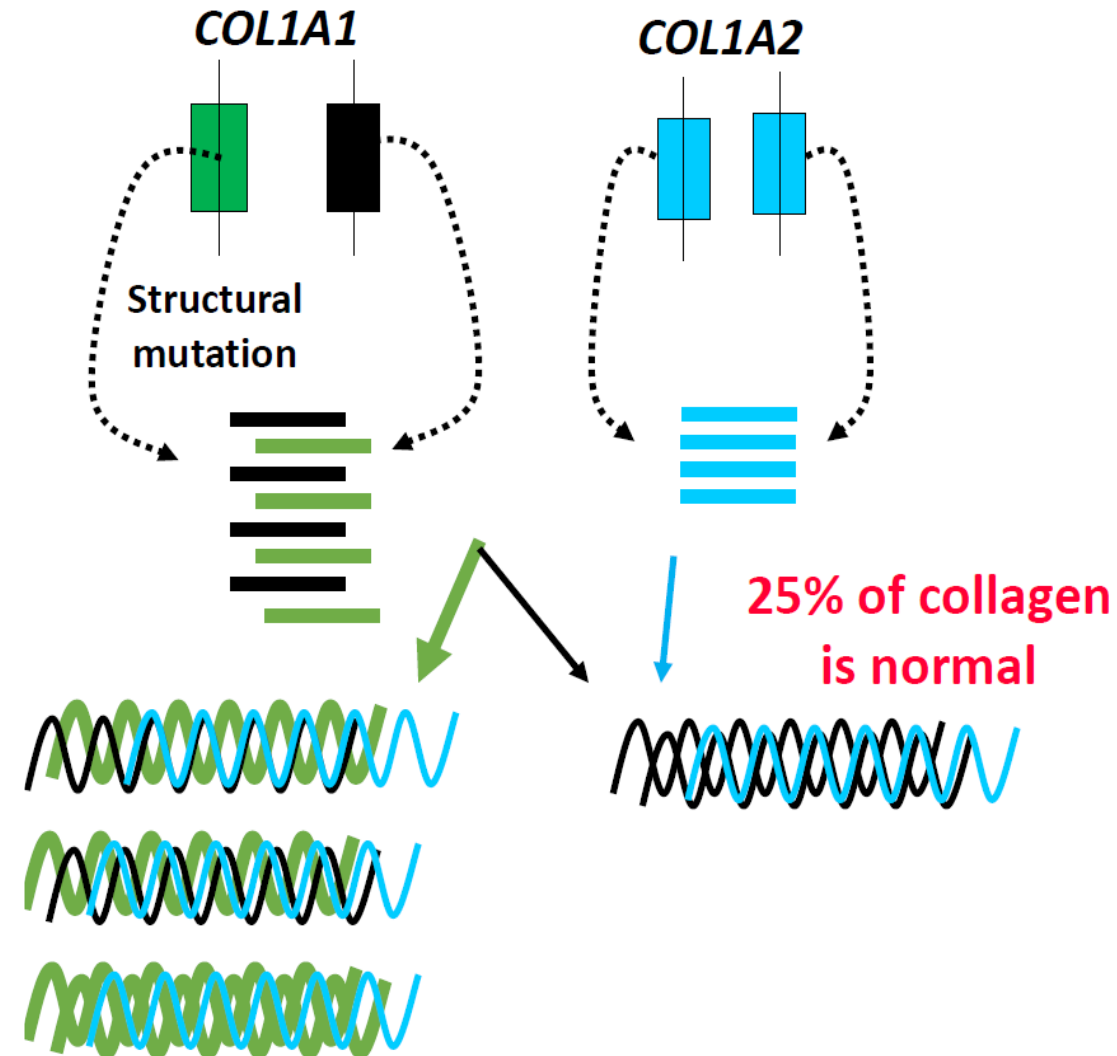
- Normal type I collagen – 2 X Col1A1 + 1 Col1A2.
- *COL1A1* expressed twice as much as *COL1A2*.
- Result = strong, properly constructed collagen!



- Loss of expression of one allele of either *COL1A1* or *COL1A2* = 50% reduction in collagen formation.
- Result = strong, properly constructed collagen but only ½ amount required.
- Haploinsufficiency (1/2 not enough!) = Classic Non-Deforming OI (Type I) with blue sclera.

Dominant Negative (Type II):

- Missense mutation (change in amino acid) may = altered structure or properties.
- Mutation can occur in either *COL1A1* or *COL1A2* gene.
- Abnormal collagen effectively disrupts normal collagen – “bad apple effect” = Dominant negative!
- Dominant negative = mechanism underlying Perinatal lethal (Type II) OI.
- Severity = level of disruption of aa change
 - Large branched chain e.g. isoleucine = Perinatal lethal
 - Smaller e.g. cysteine (FTM SG) = Classical non-deforming



Summary and Genetic Principles Illustrated:

Osteogenesis Imperfecta:

- Autosomal Dominant Inheritance
 - Haploinsufficiency
 - Dominant Negative
- Locus heterogeneity - *COL1A1* and *COL1A2* genes.
- Allelic heterogeneity – different mutation in the same gene in different families.
- Variable expressivity
 - Progressive forms to lethality
 - Environmental influence (Classic non-deforming - different exposure to physical activity)

Thank you for listening!